

PROGRAM: PAGING TECHNIQUE

C PROGRAM :

```
#include <stdio.h>
```

```
int main() {  
    int pageTable[20];  
    int pageSize;  
    int numPages;  
    int logicalAddress;  
    int pageNumber;  
    int offset;  
    int frameNumber;  
    int physicalAddress;  
    int i;  
    printf("Enter Page Size: ");  
    scanf("%d", &pageSize);  
    printf("Enter Number of Pages: ");  
    scanf("%d", &numPages);  
    printf("Enter Frame Numbers for Each Page:\n");  
    for (i = 0; i < numPages; i++) {  
        printf("Page %d -> Frame: ", i);  
        scanf("%d", &pageTable[i]);  
    }  
    printf("Enter Logical Address: ");  
    scanf("%d", &logicalAddress);  
    pageNumber = logicalAddress / pageSize;  
    offset = logicalAddress % pageSize;  
    if (pageNumber >= numPages) {  
        printf("Invalid Logical Address\n");  
        return 0;  
    }  
}
```

```

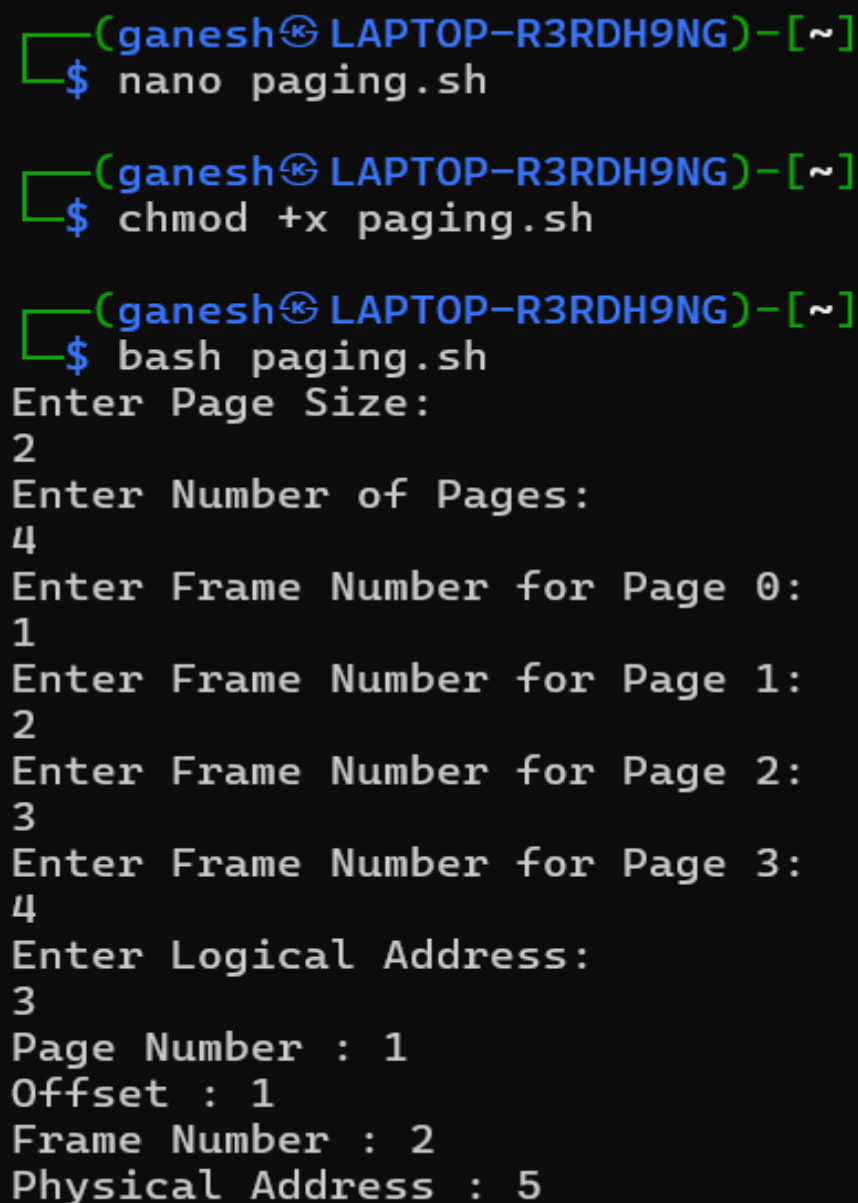
    }
    frameNumber = pageTable[pageNumber];
    physicalAddress = (frameNumber * pageSize) + offset;
    printf("\nPage Number : %d", pageNumber);
    printf("\nOffset : %d", offset);
    printf("\nFrame Number : %d", frameNumber);
    printf("\nPhysical Address : %d\n", physicalAddress);
    return 0;
}
SHELL SCRIPT :
#!/bin/bash

echo "Enter Page Size:"
read pageSize
echo "Enter Number of Pages:"
read numPages
declare -a pageTable
for ((i = 0; i < numPages; i++))
do
    echo "Enter Frame Number for Page $i:"
    read pageTable[$i]
done
echo "Enter Logical Address:"
read logicalAddress
pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))
if [ $pageNumber -ge $numPages ]
then
    echo "Invalid Logical Address"
    exit
fi

```

```
frameNumber=${pageTable[$pageNumber]}  
physicalAddress=$((frameNumber * pageSize + offset))  
echo "Page Number : $pageNumber"  
echo "Offset : $offset"  
echo "Frame Number : $frameNumber"  
echo "Physical Address : $physicalAddress"
```

Output :



```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano paging.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ chmod +x paging.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ bash paging.sh  
Enter Page Size:  
2  
Enter Number of Pages:  
4  
Enter Frame Number for Page 0:  
1  
Enter Frame Number for Page 1:  
2  
Enter Frame Number for Page 2:  
3  
Enter Frame Number for Page 3:  
4  
Enter Logical Address:  
3  
Page Number : 1  
Offset : 1  
Frame Number : 2  
Physical Address : 5
```